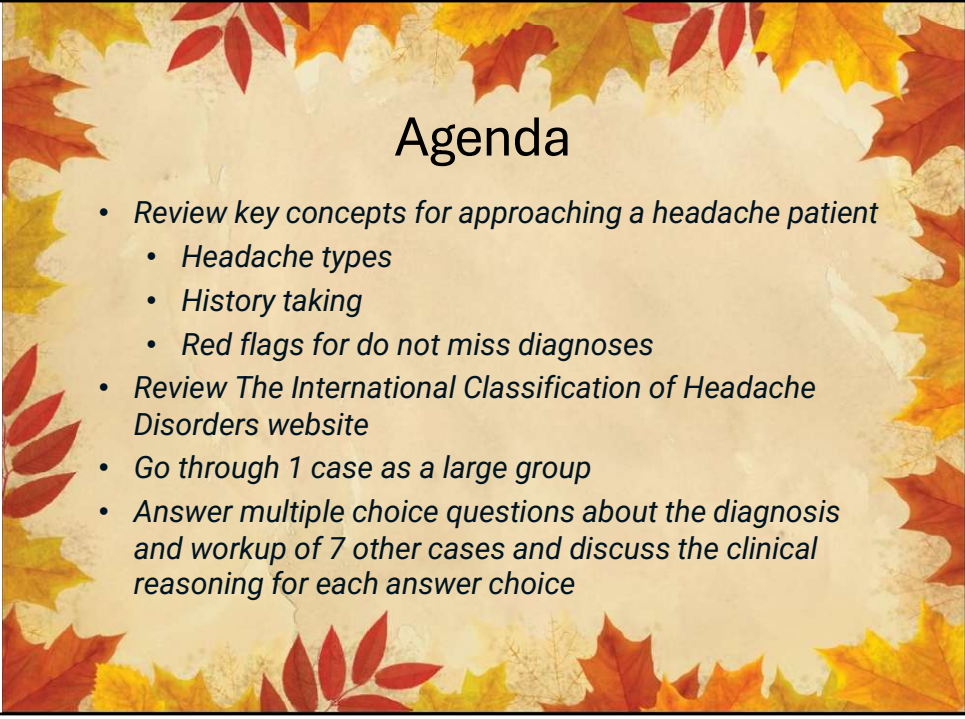


Headache DDx Session

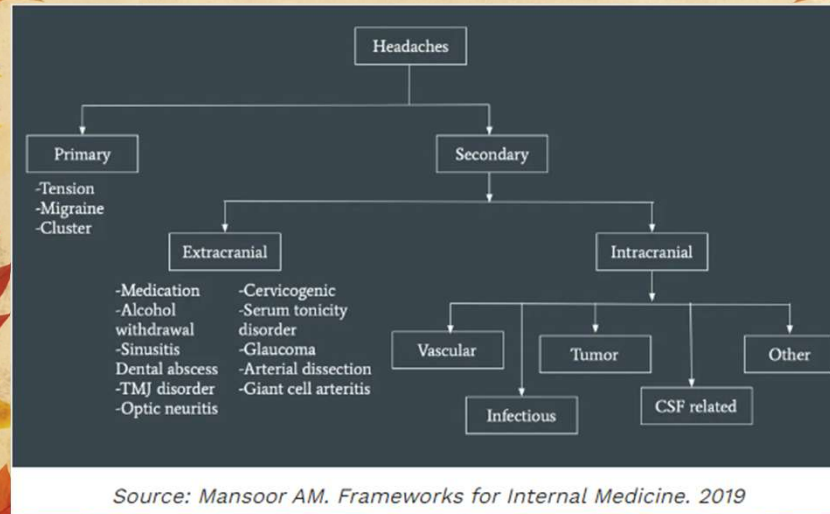
Christina Kinnevey Greig, MD



Agenda

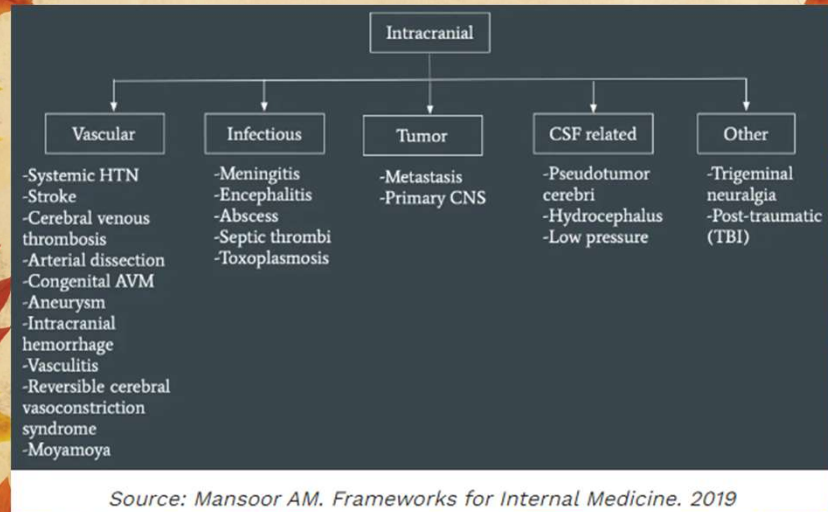
- *Review key concepts for approaching a headache patient*
 - *Headache types*
 - *History taking*
 - *Red flags for do not miss diagnoses*
- *Review The International Classification of Headache Disorders website*
- *Go through 1 case as a large group*
- *Answer multiple choice questions about the diagnosis and workup of 7 other cases and discuss the clinical reasoning for each answer choice*

Headache Types Recap



Source: Mansoor AM. Frameworks for Internal Medicine. 2019

Headaches Types Recap



Headache HPI Questions

BOLD CAARTTS	
Before	Chronic vs acute vs. recurrent. New type of headache
Onset	Sudden vs gradual, age at onset
Location	Unilateral, bilateral, retro-orbital, bitemporal
Duration	Less than 30 minutes, 30 min+, hours to days, intractable
Character	Sharp/shooting, Throbbing, Band-like tightness, Dull ache
Aggravating Factors / triggers	Position changes, valsalva, photo / phonophobia
Associated Symptoms	Neurological deficits, vision changes, numbness, tingling, lightheadedness, mental status changes, fever, nuchal rigidity, seizures, sleep disturbances
Relieving factors	Sleep, deep breathing, meditation,
Triggers	Certain foods, stress, poor sleep
Treatments tried	Medications (OTC and Rx) and other treatments
Trauma	Any recent head trauma
Severity	Typical scale from 0-10, worst ever headache

(Don't forget to also ask ROS questions in addition to the associated symptoms questions)

new onset headache or first time - onset sudden or after a
positional/Valsalva = increased intracranial pressure

What are **RED FLAGS** in each of the categories?

BOLD CAARTTTS	
Before	Chronic vs acute vs. recurrent. New type of headache
Onset	Sudden vs gradual, onset after age 50
Location	Unilateral, bilateral, retro-orbital, bitemporal
Duration	Less than 30 minutes, 30 min+, hours to days, intractable
Character	Sharp/shooting, Throbbing, Band-like tightness, Dull ache
Aggravating Factors / triggers	Position changes, valsalva , photo / phonophobia
Associated Symptoms	Neurological deficits , vision changes, numbness, tingling, lightheadedness, mental status changes, fever, nuchal rigidity, seizures, sleep disturbances
Relieving factors	Sleep, deep breathing, meditation,
Triggers	Certain foods, stress, poor sleep
Treatments tried	Medications (OTC and Rx) and other treatments
Trauma	Any recent head trauma
Severity	Typical scale from 0-10, worst ever headache

Review: Do Not Miss Diagnoses

<i>Cause</i>	<i>Symptoms</i>
Meningitis	Nuchal rigidity, headache, photophobia, and prostration; may not be febrile. Lumbar puncture is diagnostic.
Intra-cranial hemorrhage	Nuchal rigidity and headache; may not have clouded consciousness or seizures. Hemorrhage may not be seen on CT scan. Lumbar puncture shows "bloody tap" that does not clear by the last tube. A fresh hemorrhage may not be xanthochromic.
Brain tumor	May present with prostrating pounding headaches that are associated with nausea and vomiting. Should be suspected in progressively severe new "migraine" that is invariably unilateral.
Temporal arteritis	May present with a unilateral pounding headache. Onset generally in older patients (>50 years) and frequently associated with visual changes. The erythrocyte sedimentation rate is the best screening test and is usually markedly elevated (i.e., >50). Definitive diagnosis can be made by arterial biopsy.
Glaucoma	Usually consists of severe eye pain. May have nausea and vomiting. The eye is usually painful and red. The pupil may be partially dilated.

Specific features suggesting a secondary headache source — Other features that suggest a specific source of headache pain include the following:

- Impaired vision or seeing halos around light suggests the presence of glaucoma. Suspicion for subacute angle closure glaucoma should be raised by relatively short duration (often less than one hour) unilateral headaches that do not meet criteria for migraine arising after age 50 [16].
- Visual field defects suggest the presence of a lesion of the optic pathway (eg, due to a pituitary mass).
- Sudden, severe, unilateral vision loss suggests the presence of optic neuritis. Optic neuritis typically presents with painful, monocular visual loss that evolves over several hours to a few days. One-third of patients have visible optic nerve inflammation (papillitis) on fundoscopic examination. (See ["Optic neuritis: Pathophysiology, clinical features, and diagnosis"](#).)
- Blurring of vision on forward bending of the head, headaches upon waking early in the morning that improve with sitting up, and double vision or loss of coordination and balance should raise the suspicion of raised intracranial pressure (ICP); this should also be considered in patients with chronic, daily, progressively worsening headaches associated with chronic nausea. Idiopathic

intracranial hypertension (pseudotumor cerebri) typically affects obese women of child-bearing age. Characteristic features are headache, papilledema, vision loss or diplopia, elevated lumbar puncture (LP) opening pressure with normal cerebrospinal fluid (CSF) composition. (See ["Evaluation and management of elevated intracranial pressure in adults"](#) and ["Idiopathic intracranial hypertension \(pseudotumor cerebri\): Clinical features and diagnosis"](#).)

●In patients who present with headache that is relieved with recumbency and exacerbated with upright posture, the diagnosis of headache attributed to spontaneous intracranial hypotension should be considered. An additional major feature of this headache syndrome is diffuse meningeal enhancement on brain MRI. The accepted etiology is CSF leakage, which may occur in the context of rupture of an arachnoid membrane. (See ["Spontaneous intracranial hypotension: Pathophysiology, clinical features, and diagnosis"](#).)

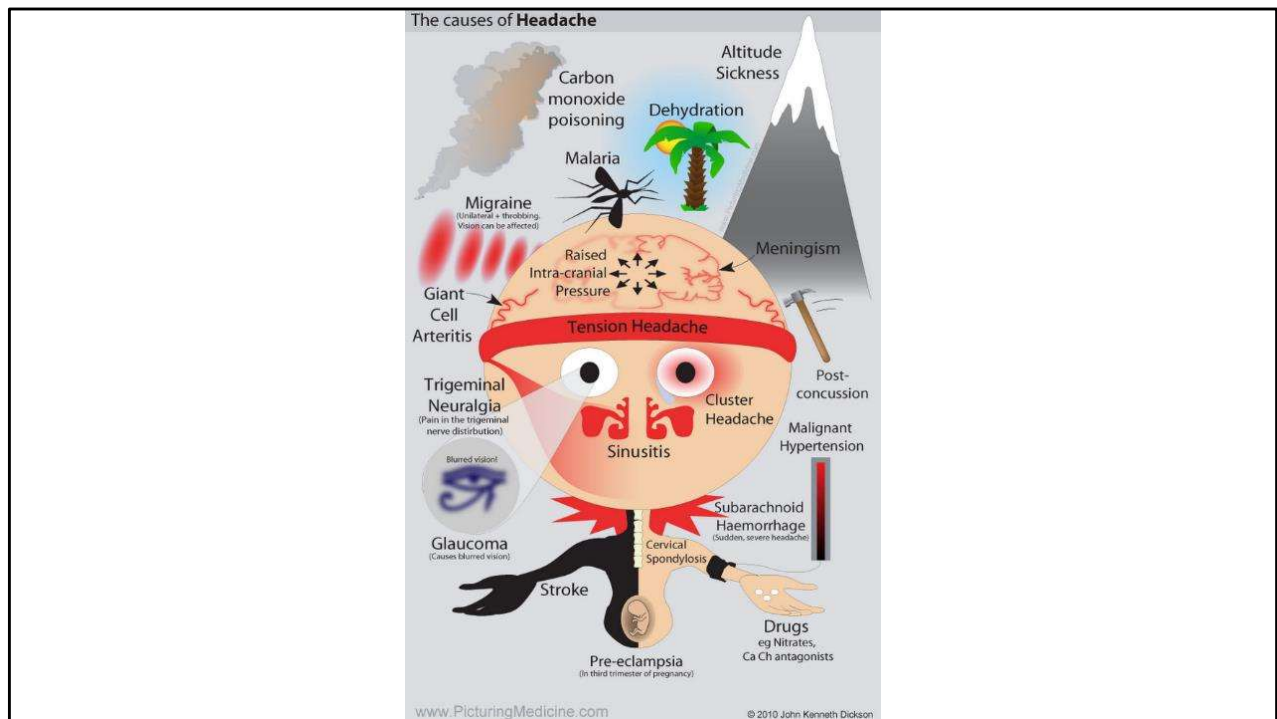
●The presence of nausea, vomiting, worsening of headache with changes in body position (particularly bending over), an abnormal neurologic examination, and/or a significant change in prior headache pattern suggests the headache was caused by a tumor. (See ["Overview of the clinical features and diagnosis of brain tumors in adults"](#).)

●Intermittent headache with generalized sweating, tachycardia, and/or sustained or paroxysmal hypertension is suggestive of pheochromocytoma. (See ["Clinical presentation and diagnosis of pheochromocytoma"](#).)

●Morning headache is nonspecific and can occur as part of a primary headache syndrome or may be secondary to a number of disorders including sleep apnea, chronic obstructive pulmonary disease, and the obesity hypoventilation syndrome. (See ["Clinical presentation and diagnosis of obstructive sleep apnea in adults"](#) and ["Chronic obstructive pulmonary disease: Definition, clinical manifestations, diagnosis, and staging"](#) and ["Clinical manifestations and diagnosis of obesity hypoventilation syndrome"](#).)



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